

Boss Challenge — Paper 43 (Class 7)

Physical & Chemical Changes | Reproduction in Plants | Fractions & Decimals | The Triangle & its Properties
One correct option each. Marking: +4 correct, -1 wrong, 0 blank. Write A/B/C/D on a separate sheet.

1. A change in which no new substance is formed and which can usually be reversed is called a:
(A) chemical change
(B) permanent change
(C) nuclear change
(D) physical change
2. The making of new individuals of the same kind by living things, so the species continues, is called:
(A) photosynthesis
(B) reproduction
(C) germination
(D) respiration
3. Multiplying the two fractions $\frac{1}{2}$ and $\frac{1}{3}$ together gives a product equal to:
(A) $\frac{1}{5}$
(B) $\frac{1}{6}$
(C) $\frac{2}{6}$
(D) $\frac{2}{5}$
4. In any triangle, the three angles measured inside it together add up to exactly:
(A) 180°
(B) 360°
(C) 90°
(D) 270°
5. Ice melting into water and water freezing back into ice are classic examples of a change that is:
(A) physical but irreversible
(B) chemical and reversible
(C) physical and reversible
(D) chemical and irreversible
6. Reproduction in which a new plant grows from a part of the parent plant, without seeds, is called:
(A) pollination
(B) fertilisation
(C) sexual reproduction
(D) vegetative propagation
7. The product $\frac{3}{4} \times 8$ is equal to:
(A) 2
(B) 11
(C) 24
(D) 6

8. If a triangle's first two angles measure 50° and 60° , then its remaining angle must be:
- (A) 70°
 - (B) 110°
 - (C) 60°
 - (D) 80°
9. When iron is left in damp air it slowly turns into a reddish-brown flaky layer; this rust is a sign of a:
- (A) chemical change
 - (B) a reversible change
 - (C) change of state only
 - (D) physical change
10. The small bud-like outgrowths on the leaf of a Bryophyllum plant are special because each one can:
- (A) make food for the leaf
 - (B) store extra water
 - (C) grow into a new plant
 - (D) turn into flowers only
11. Among the fractions $\frac{1}{2}$, $\frac{2}{3}$, $\frac{3}{4}$ and $\frac{5}{12}$, the one with the greatest value is:
- (A) $\frac{3}{4}$
 - (B) $\frac{5}{12}$
 - (C) $\frac{2}{3}$
 - (D) $\frac{1}{2}$
12. When every one of a triangle's three sides has exactly the same length, we call it:
- (A) an equilateral triangle
 - (B) a scalene triangle
 - (C) a right triangle
 - (D) an isosceles triangle
13. A nail of mass 8.0 g is left out and gains rust until it weighs 8.6 g. The extra 0.6 g comes mainly from the:
- (A) water it dried out
 - (B) dust settling on it
 - (C) iron escaping into the air
 - (D) oxygen it joined with from the air
14. Tiny plants such as yeast reproduce by forming a small bulge that grows and breaks away. This method is called:
- (A) pollination
 - (B) fragmentation
 - (C) germination
 - (D) budding

15. The decimal 0.75 written as a fraction in its simplest form is:

- (A) $\frac{3}{4}$
- (B) $\frac{3}{5}$
- (C) $\frac{7}{5}$
- (D) $\frac{75}{10}$

16. A triangle that has exactly two equal sides is called:

- (A) a scalene triangle
- (B) an equilateral triangle
- (C) an isosceles triangle
- (D) a right triangle

17. Magnesium ribbon burns with a dazzling white light and leaves a white ash. The dazzling flame plus the new ash tell us this is a:

- (A) chemical change
- (B) change of state
- (C) physical change
- (D) reversible change

18. Some simple water plants like Spirogyra break into two or more pieces, and each piece grows into a new plant. This is called:

- (A) budding
- (B) fertilisation
- (C) fragmentation
- (D) pollination

19. The reciprocal of the fraction $\frac{2}{5}$ is:

- (A) $\frac{2}{5}$
- (B) $\frac{7}{10}$
- (C) $\frac{5}{2}$
- (D) $-\frac{2}{5}$

20. When no two sides of a triangle share the same length at all, the triangle is called:

- (A) an acute triangle
- (B) an isosceles triangle
- (C) a scalene triangle
- (D) an equilateral triangle

21. Three-quarters of a 20 cm magnesium ribbon is burnt away. The length of ribbon that turned to ash is:

- (A) 12 cm
- (B) 15 cm
- (C) 5 cm
- (D) 16 cm

22. The tiny, light, dust-like reproductive units made in large numbers by ferns and mosses are called:

- (A) buds
- (B) fruits
- (C) seeds
- (D) spores

23. Dividing the fraction $\frac{1}{2}$ by the fraction $\frac{1}{4}$ gives an answer of:

- (A) 8
- (B) 2
- (C) $\frac{1}{2}$
- (D) $\frac{1}{8}$

24. A triangle that contains one right angle of 90° is called:

- (A) an obtuse triangle
- (B) an equilateral triangle
- (C) a right-angled triangle
- (D) an acute triangle

25. Burning a candle involves both kinds of change at once: the wax melting near the flame is physical, while the wax burning to give light and heat is:

- (A) also a physical change
- (B) a change of state only
- (C) a chemical change
- (D) no change at all

26. In a flower, the part that makes pollen grains and is the male reproductive part is the:

- (A) petal
- (B) stamen
- (C) sepal
- (D) pistil

27. The sum $0.6 + 0.45$ is equal to:

- (A) 0.105
- (B) 0.51
- (C) 1.05
- (D) 1.5

28. An exterior angle of a triangle equals the sum of the two:

- (A) adjacent interior angles
- (B) base angles only
- (C) interior opposite angles
- (D) right angles in the figure

29. When you mix baking soda with vinegar it fizzes and bubbles of gas rush out. The fizzing tells you that a:

- (A) the liquid is only getting colder
- (B) the soda is simply dissolving
- (C) new gas is being formed
- (D) the vinegar is evaporating

30. The female reproductive part of a flower, which includes the stigma, style and ovary, is the:

- (A) anther
- (B) pistil
- (C) stamen
- (D) petal

31. The product 0.2×0.3 is equal to:

- (A) 0.6
- (B) 0.06
- (C) 6
- (D) 0.5

32. In a triangle, an exterior angle is 120° and one interior opposite angle is 70° . The other interior opposite angle is:

- (A) 120°
- (B) 50°
- (C) 60°
- (D) 190°

33. Curd setting from milk, food being cooked, and a leaf rotting are alike because each one is a:

- (A) chemical change
- (B) physical change
- (C) reversible change
- (D) change of state only

34. Carrying pollen grains from a flower's anther across to a stigma is the process known as:

- (A) pollination
- (B) fertilisation
- (C) germination
- (D) dispersal

35. The value of $2.5 \div 0.5$ is:

- (A) 12.5
- (B) 1.25
- (C) 5
- (D) 0.5

36. The longest side of a right-angled triangle, which lies opposite the right angle, is called the:
- (A) median
 - (B) hypotenuse
 - (C) base
 - (D) altitude
37. Galvanisation protects iron from rusting by coating it with a thin layer of:
- (A) zinc
 - (B) copper
 - (C) lead
 - (D) gold
38. When pollen lands on the stigma of a flower of the SAME plant, it is called:
- (A) cross-pollination
 - (B) fertilisation
 - (C) germination
 - (D) self-pollination
39. In the fraction $\frac{7}{9}$, the number 9 below the line is called the:
- (A) denominator
 - (B) numerator
 - (C) quotient
 - (D) reciprocal
40. By the Pythagoras property, in a right triangle with legs 3 and 4, the hypotenuse is:
- (A) 12
 - (B) 5
 - (C) 7
 - (D) 25
41. Dissolving sugar in water is called a physical change mainly because the sugar:
- (A) changes into water itself
 - (B) can be recovered by evaporating the water
 - (C) can never be got back again
 - (D) turns into a new sweet substance
42. The joining of the male cell from the pollen with the female cell in the ovule, which forms the seed, is called:
- (A) pollination
 - (B) germination
 - (C) fertilisation
 - (D) budding

43. A recipe needs $\frac{3}{4}$ cup of flour for one cake. For 3 cakes you need:
- (A) $1\frac{1}{4}$ cups
 - (B) $\frac{3}{4}$ cup
 - (C) $2\frac{1}{4}$ cups
 - (D) $\frac{9}{4}$ cups left as is
44. A line segment joining a vertex of a triangle to the midpoint of the opposite side is called a:
- (A) hypotenuse
 - (B) base
 - (C) altitude
 - (D) median
45. A 2.5 g spoon of salt is stirred into water and then the water is fully evaporated. The mass of salt you get back is:
- (A) 2.5 g
 - (B) 0 g
 - (C) 1.25 g
 - (D) 5 g
46. In a field, 4 out of every 5 sown seeds sprout. The fraction of seeds that sprout is:
- (A) $\frac{5}{4}$
 - (B) $\frac{4}{5}$
 - (C) $\frac{4}{9}$
 - (D) $\frac{1}{5}$
47. The fraction $\frac{12}{16}$ written in its simplest form is:
- (A) $\frac{3}{4}$
 - (B) $\frac{6}{8}$
 - (C) $\frac{4}{3}$
 - (D) $\frac{12}{16}$
48. The perpendicular line segment drawn from a vertex to the opposite side of a triangle is the:
- (A) hypotenuse
 - (B) median
 - (C) exterior angle
 - (D) altitude
49. When crystals of a substance are formed slowly from its hot solution, the process is called:
- (A) galvanisation
 - (B) condensation
 - (C) crystallisation
 - (D) evaporation

50. A plant produces 200 seeds and $\frac{3}{4}$ of them are eaten by birds. The number of seeds eaten is:
- (A) 75
 - (B) 150
 - (C) 50
 - (D) 100
51. Half of $\frac{1}{2}$ is:
- (A) $\frac{1}{2}$
 - (B) $\frac{2}{2}$
 - (C) $\frac{1}{4}$
 - (D) 1
52. Which set of side lengths can actually form a triangle?
- (A) 1 cm, 2 cm, 5 cm
 - (B) 1 cm, 1 cm, 3 cm
 - (C) 2 cm, 2 cm, 5 cm
 - (D) 3 cm, 4 cm, 6 cm
53. Souring of milk, rusting of iron and burning of wood are usually hard to reverse. This tells us that chemical changes are often:
- (A) permanent
 - (B) changes of state
 - (C) easily reversed
 - (D) temporary
54. Seeds with wings or hair, such as those of the maple or the drumstick, are mainly spread by:
- (A) wind
 - (B) animals
 - (C) water
 - (D) exploding fruits
55. Which decimal is equal to the fraction $\frac{1}{5}$?
- (A) 1.5
 - (B) 0.5
 - (C) 0.2
 - (D) 0.15
56. In an isosceles triangle the two equal sides each meet the base, and the two base angles are:
- (A) always different
 - (B) always right angles
 - (C) equal to each other
 - (D) always 60° each

57. Ozone protects us from the Sun's harmful rays. Its formation from oxygen high in the air is a chemical change because:

- (A) nothing new is made
- (B) the air simply gets warmer
- (C) the oxygen only changes state
- (D) a new substance, ozone, is formed

58. Seeds that have hooks or a sticky surface, like those of Xanthium, are usually carried away by:

- (A) self-bursting
- (B) wind
- (C) water
- (D) animals

59. The value of $\frac{5}{8} - \frac{1}{8}$ is:

- (A) $\frac{4}{16}$
- (B) $\frac{4}{8}$ only
- (C) $\frac{6}{8}$
- (D) $\frac{1}{2}$

60. In an equilateral triangle, where all the sides match, every single one of its angles measures:

- (A) 45°
- (B) 90°
- (C) 120°
- (D) 60°

61. A blacksmith heats an iron rod until it glows and then hammers it into a curved shape. Bending the softened iron is a:

- (A) physical change
- (B) crystallisation
- (C) chemical change
- (D) rusting

62. Spreading seeds far away from the parent plant is helpful mainly because it:

- (A) turns the seeds into flowers
- (B) makes the seeds heavier
- (C) reduces crowding and competition for water, minerals and sunlight
- (D) stops the seeds from sprouting

63. A wire 2.4 m long is cut into 4 equal pieces. Each piece is:

- (A) 0.06 m
- (B) 1.0 m
- (C) 0.8 m
- (D) 0.6 m

64. Is it possible for one single triangle to contain two separate right angles of 90° each?
- (A) No, because the three angles would add up to more than 180°
 - (B) Yes, if the sides are very long
 - (C) Yes, in a right-angled triangle
 - (D) Yes, if it is equilateral
65. To slow the rusting of an iron chain you should mainly keep it:
- (A) dry and away from moisture
 - (B) wet at all times
 - (C) painted with salty water
 - (D) buried in damp soil
66. After fertilisation, the ovary of the flower grows and ripens into a:
- (A) a root
 - (B) a leaf
 - (C) a new flower
 - (D) fruit
67. Comparing 0.7 and 0.65, the larger number is:
- (A) they are equal
 - (B) 0.7
 - (C) cannot be compared
 - (D) 0.65
68. A right-angled triangle has legs 6 cm and 8 cm. The length of the hypotenuse is:
- (A) 14 cm
 - (B) 100 cm
 - (C) 48 cm
 - (D) 10 cm
69. Photosynthesis, in which a plant turns carbon dioxide and water into food using sunlight, is a chemical change because the plant:
- (A) simply warms the water inside
 - (B) makes a new substance, sugar
 - (C) only changes the colour of light
 - (D) freezes the carbon dioxide
70. Gardeners often grow a new rose plant by planting a cut piece of its stem. This piece is called a:
- (A) spore
 - (B) bud
 - (C) cutting
 - (D) seed

71. Rewriting the improper fraction $7/2$ in the form of a mixed number gives:
- (A) $1 \frac{3}{2}$
 - (B) $2 \frac{1}{3}$
 - (C) $3 \frac{1}{3}$
 - (D) $3 \frac{1}{2}$
72. Two angles of a triangle measure 90° and 45° . Such a triangle is BOTH:
- (A) right-angled and isosceles
 - (B) equilateral and acute
 - (C) right-angled and equilateral
 - (D) obtuse and scalene
73. A piece of paper torn into many small bits is still paper. Tearing paper is therefore a:
- (A) burning
 - (B) chemical change
 - (C) physical change
 - (D) an irreversible chemical change
74. Bright petals, sweet smell and nectar in a flower are mainly there to:
- (A) frighten away all animals
 - (B) make food for the plant
 - (C) store water for dry days
 - (D) attract insects that help in pollination
75. One-third of 0.9 is:
- (A) 0.27
 - (B) 0.3
 - (C) 3
 - (D) 0.6
76. A roof is a triangle with a base of 8 m and a height of 3 m. Its area is:
- (A) 6 m^2
 - (B) 12 m^2
 - (C) 24 m^2
 - (D) 11 m^2
77. In one experiment, 0.4 g of a metal reacts and another 0.4 g reacts the next day, while 0.2 g stays unreacted. The total mass of metal that reacted is:
- (A) 0.2 g
 - (B) 0.4 g
 - (C) 0.8 g
 - (D) 1.0 g

78. A pod of a balsam plant suddenly bursts open and flings its seeds out. This is dispersal by:

- (A) sticky hooks
- (B) water only
- (C) wind only
- (D) the bursting of the fruit itself

79. The product $\frac{2}{3} \times \frac{9}{4}$ in its simplest form is:

- (A) $\frac{2}{3}$
- (B) $\frac{3}{2}$
- (C) $\frac{11}{7}$
- (D) $\frac{18}{12}$ left as is

80. In a triangle the angles are in the ratio 1 : 2 : 3. The largest angle is:

- (A) 60°
- (B) 120°
- (C) 30°
- (D) 90°

81. Setting of cement after water is added is a chemical change. The strongest clue is that the hardened cement:

- (A) can easily be made into powder again
- (B) looks a little greyer than before
- (C) cannot be turned back into soft cement powder
- (D) feels slightly cooler to touch

82. A seed is made up of a tiny baby plant called the embryo together with a store of:

- (A) water for the soil
- (B) food for the young plant
- (C) salt for the roots
- (D) pollen for the flower

83. Rohan reads $\frac{1}{4}$ of a book on Monday and $\frac{2}{4}$ of it on Tuesday. The fraction of the book he has read is:

- (A) $\frac{2}{4}$
- (B) $\frac{3}{4}$
- (C) $\frac{3}{8}$
- (D) $\frac{1}{2}$

84. A triangular garden has sides 5 m, 7 m and 9 m. The length of fencing needed to go all around it is:

- (A) 21 m
- (B) 315 m
- (C) 63 m
- (D) 12 m

85. Melting of wax, boiling of water and dissolving of sugar are grouped together because each one is a:
- (A) physical change
 - (B) a change that makes a new substance
 - (C) irreversible change
 - (D) chemical change
86. A new plant grown by vegetative propagation, such as from a potato eye, is exactly like the parent because it:
- (A) mixes features from two parents
 - (B) makes its own brand-new flowers first
 - (C) comes from a single parent and has the same features
 - (D) always grows much larger than the parent
87. Multiplying the decimal number 1.2 by the whole number 5 gives a result of:
- (A) 0.6
 - (B) 5.2
 - (C) 6.0
 - (D) 60
88. A triangle has one angle of 100° . This triangle is:
- (A) an equilateral triangle
 - (B) an acute triangle
 - (C) a right triangle
 - (D) an obtuse triangle
89. Rusting is faster near the sea. The main reason is that the salty sea air supplies extra:
- (A) sunlight that melts the iron
 - (B) moisture and salt that speed the reaction
 - (C) sand that scratches the iron
 - (D) cool air that freezes the iron
90. A mango tree drops 60 seeds and only $\frac{1}{3}$ of them land on good soil. The number that land on good soil is:
- (A) 20
 - (B) 30
 - (C) 3
 - (D) 40
91. Which fraction is equivalent to $\frac{2}{3}$?
- (A) $\frac{8}{12}$
 - (B) $\frac{6}{8}$
 - (C) $\frac{4}{9}$
 - (D) $\frac{3}{2}$

- 92.** The three medians of a triangle always meet at a single point called the:
- (A) centroid
 - (B) hypotenuse
 - (C) vertex
 - (D) right angle
- 93.** A burnt matchstick will not light again, but a melted ice cube can be re-frozen. This contrast shows that, in general:
- (A) chemical changes are hard to reverse while many physical changes are not
 - (B) chemical changes are always easy to reverse
 - (C) both kinds of change are equally easy to reverse
 - (D) physical changes are always permanent
- 94.** Out of 100 flowers on a tree, 0.25 of them turn into fruit. The number of flowers that become fruit is:
- (A) 4
 - (B) 75
 - (C) 50
 - (D) 25
- 95.** The number 0.125 written as a fraction in simplest form is:
- (A) $\frac{125}{100}$
 - (B) $\frac{1}{8}$
 - (C) $\frac{1}{4}$
 - (D) $\frac{1}{5}$
- 96.** In an isosceles triangle each of the two equal base angles is 65° . The third angle (the vertex angle) is:
- (A) 115°
 - (B) 50°
 - (C) 130°
 - (D) 65°
- 97.** Digesting food in your body, where it is broken down into new, simpler substances your cells can use, is best described as a:
- (A) change of state only
 - (B) chemical change
 - (C) a reversible change
 - (D) physical change
- 98.** Reproduction by seeds, which needs pollen and an egg from flowers, is called sexual reproduction because it involves:
- (A) breaking the plant into pieces
 - (B) only one parent's body part
 - (C) a bud pinching off the parent
 - (D) the joining of two different cells, male and female

99. A jug holds 1.5 litre and a glass holds 0.25 litre. The number of full glasses the jug can fill is:

- (A) 4
- (B) 3
- (C) 1.75
- (D) 6

100. A triangle with all three angles less than 90° is called:

- (A) a right triangle
- (B) an acute triangle
- (C) an obtuse triangle
- (D) a scalene triangle